Yuxuan Zhu

AI for Security & Statistically Rigorous Data Systems for Analytics and AI
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RESEARCH INTERESTS

- Applying AI methods to strengthen cybersecurity attacks and defenses.
- Statistically rigorous evaluation of AI/LLM agents.
- Efficient, theoretically sound systems for data analytics.

EDUCATION

University of Illinois Urbana-Champaign 2023 — present, exp. 2028
Ph.D. in computer science
Advisor: Daniel Kang

University of Michigan 2021 — 2023
Bachelor of Science and Engineering, Computer Science

Shanghai Jiao Tong University 2019 — 2023
Bachelor of Science, Electrical and Computer Engineering

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign 2023 — present

- Built **CVE-Bench**, first benchmark evaluating AI agents' ability to exploit 40 real-world severe CVEs; adopted by US AI Safety Institute.
- Designed **PilotDB**, online database-agnostic approximate query processing system with statistical guarantees; $> 5\times$ speedups on TPC-H with $\leq 5\%$ error.
- Assessing statistical rigor of AI agent benchmarks (first author, under review).
- Multi-agent framework that autonomously exploits zero-day web vulnerabilities (first author, under review).
- Approximate join processing over unstructured data (first author, under review).

Symbiotic Lab, University of Michigan (Advised by Fan Lai and Prof. Mosharaf Chowdhury.) 2022 — 2023

- Built **FedTrans**, a personalized on-device distributed ML training system.

Intelligent Programming Lab, University of Michigan (Advised by Prof. Xinyu Wang.) 2021 — 2023

- Developed **SlabCity**, a whole-query optimization system leveraging program synthesis.

SELECTED PUBLICATIONS

- **Y. Zhu**, A. Kellermann, A. Gupta, P. Li, R. Fang, R. Bindu, D. Kang. "Teams of LLM Agents Can Exploit Zero-Day Vulnerabilities" *Preprint* 2025.
- **Y. Zhu**, A. Kellermann, D. Bowman, D. Kang, and others. "CVE-Bench: A Benchmark for AI Agents' Ability to Exploit Real-World Web Application Vulnerabilities" *ICML* 2025 Spotlight.
- **Y. Zhu**, T. Jin, S. Baziotis, C. Zhang, C. Mendis, D. Kang. "Database-Agnostic Online Approximate Query Processing with A Priori Error Guarantees" *SIGMOD* 2025.
- **Y. Zhu**, D. Kang. "Efficient Approximate Query Processing with Block Sampling" *CIDR* 2025.
- **Y. Zhu**, J. Liu, F. Lai, M. Chowdhury. "FedTrans: Efficient Federated Learning via Multi-Model Transformation" *MLSys* 2024.
- R. Dong, J. Liu, **Y. Zhu**, C. Yan, B. Mozafari, X. Wang. "SlabCity: Whole-Query Optimization using Program Synthesis " *PVLDB* 2023.

SELECTED HONORS

- **2nd Prize**, SafeBench competition. 2025
- **Winner**, Microsoft Student Hackathon - Hack for Education (AI&ML). 2021
- Undergraduate awards: Tang Junyuan Scholarship, Univ. Physics Comp. Bronze, Academic Scholarship. 2020 — 2021

TEACHING & SERVICE

- Artifact reviewer, SIGMOD 2024
- Grader, MATH 417 Matrix Algebra, University of Michigan 2022